

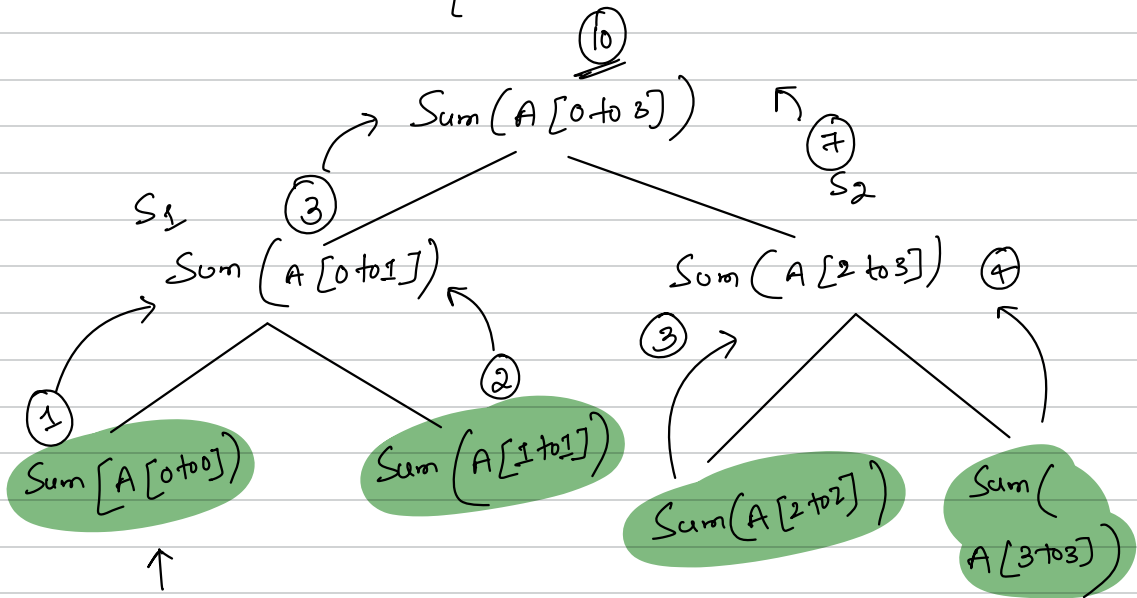


Recursion

Recursion in Computer Science is a method where Solutions to problem depends on Solutions to Smaller instance of the Same problem.

Find Sum of the Array.

$$A = \overset{0}{1}, \overset{1}{2}, \overset{2}{3}, \overset{3}{4}$$

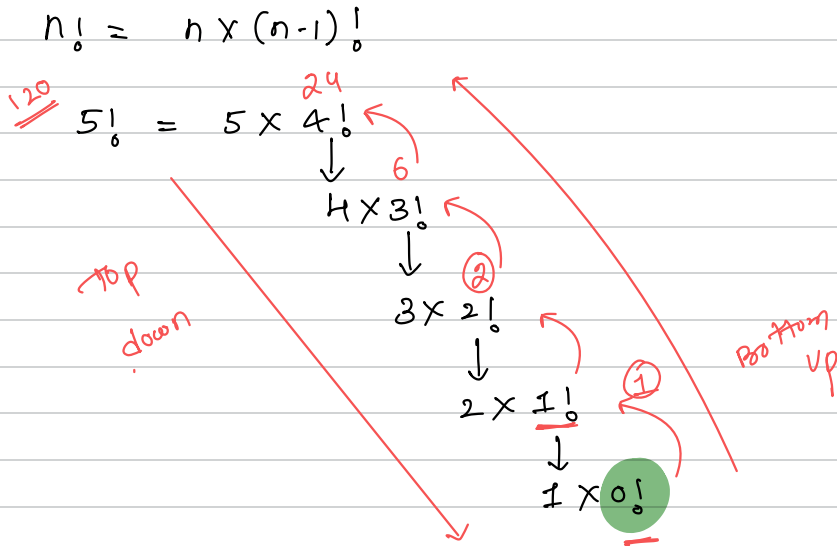


to find the Sum of the Entire Array
it look big problem So We break into
Smaller Sub problems.

Parts of Recursion Algorithm

① Base Case

② Recursive call (will work towards Base Case)



$$f(n) = \begin{cases} n \times f(n-1) \rightarrow \text{Recursive call} \\ f(0) = 1 \quad \text{Base Case} \end{cases}$$

Easiest way to Approach Recursion problem:

Magical Recursion Rule:-

Principle of Mathematical Induction

[PMI]

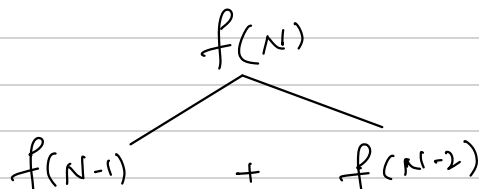
① figure out the base

② Assume that your sub problems were solved by recursion (automatically)

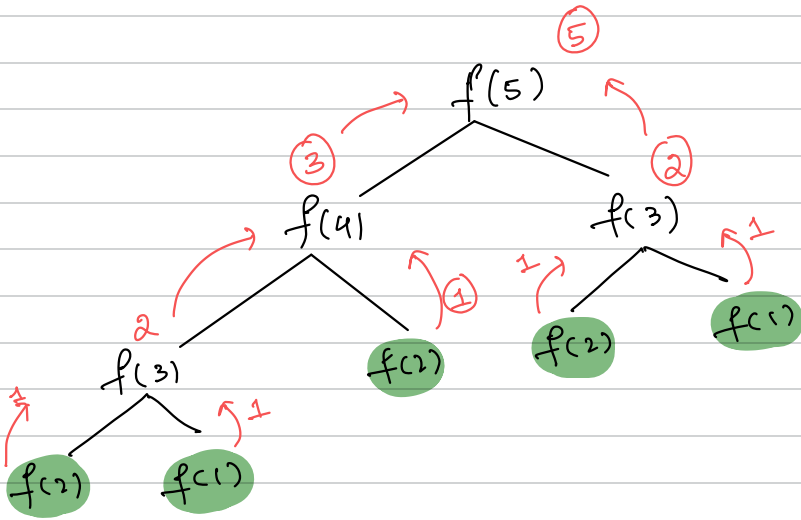
③ Using the sub problems answers write the curr problem solution.

* problem : find n^{th} fibonacci Using Recursion.

0 1 1 2 3 5 8 13 . . .



$$f(N) = \begin{cases} f(N-1) + f(N-2) \Rightarrow \text{Recursive call} \\ f(1) = 1, f(2) = 1 \Rightarrow \text{Base case} \end{cases}$$



Any Recursion problem have two things
main

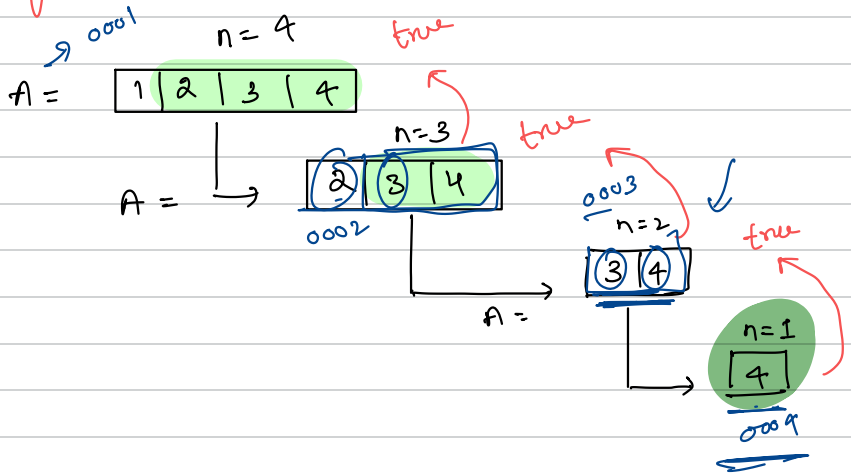
① Thinking

└→ base
└→ Recursion call

② Implementation

Problem

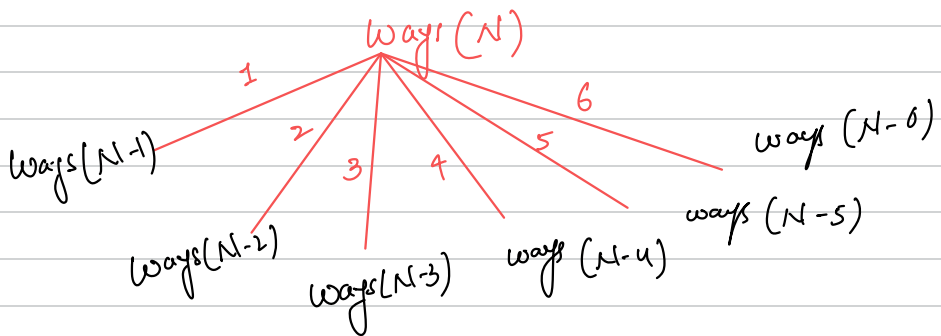
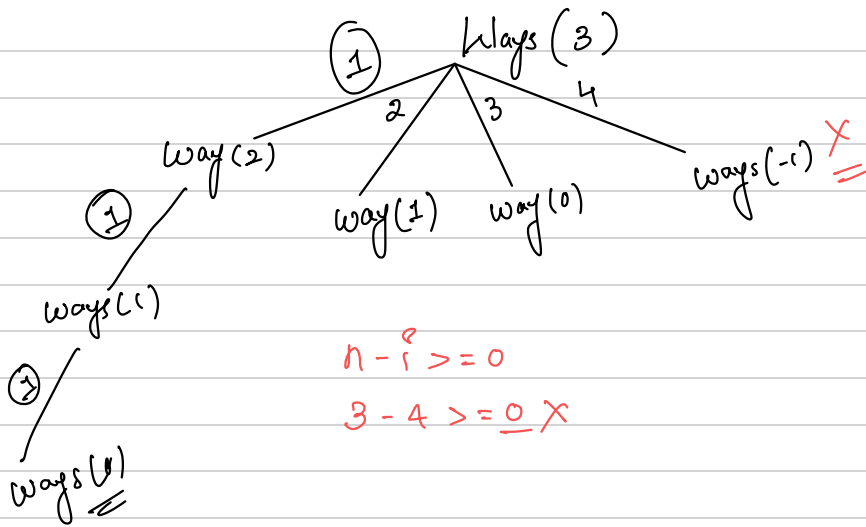
Write a recursive function to check the
Given array is Sorted or not



Dice Combinations

$n=3$

$$\begin{array}{lcl} 1 + 1 + 1 & = & 3 \\ 1 + 2 & = & 3 \\ 2 + 1 & = & 3 \\ 3 & = & 3 \end{array} \quad \left. \vphantom{\begin{array}{lcl} 1 + 1 + 1 \\ 1 + 2 \\ 2 + 1 \\ 3 \end{array}} \right\} \textcircled{4}$$



$$\text{ways}(N) = \text{ways}(N-1) + \text{ways}(N-2) + \dots$$

$$\text{ways}(N-6)$$

$$= \sum_{i=1}^6 \text{ways}(N-i)$$

Subsets & Permutation

$$A = \overset{0}{\cancel{1}}, \overset{1}{2}, \overset{2}{3} \quad (3)$$

$\{\}$

$\{1\}$

$\{1, 2\}$

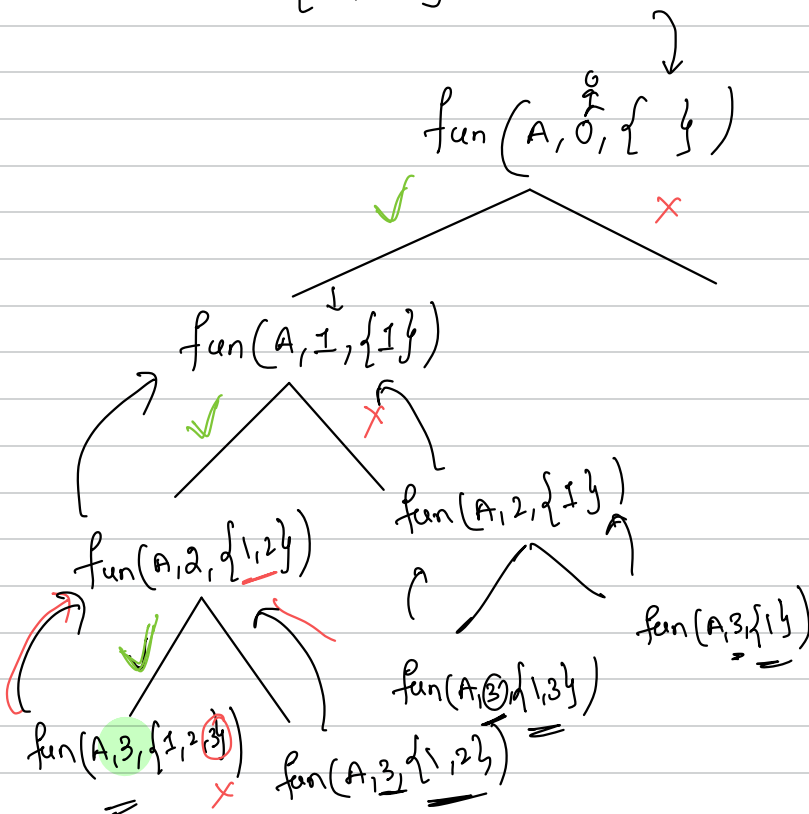
$\{2, 3\}$

$\{2\}$

$\{1, 3\}$

$\{3\}$

$\{1, 2, 3\}$



$$A = \{1, 2, 3\}$$

$$\{1, 2, 3\}$$

$$\{3, 1, 2\}$$

$$\{1, 3, 2\}$$

$$\{3, 2, 1\}$$

$$\{2, 1, 3\}$$

$$\{2, 3, 1\}$$

